

1. Evaluate $\int_{-2}^3 x^{-4} dx$ or show that it cannot be evaluated.

Solution.

$$\int_{-2}^3 x^{-4} dx = \lim_{r \rightarrow 0^-} \int_{-2}^r x^{-4} dx + \lim_{s \rightarrow 0^+} \int_s^3 x^{-4} dx$$

if the limits exist.

$$\begin{aligned} \int_{-2}^r x^{-4} dx &= \left. -\frac{1}{3x^3} \right]_{-2}^r = -\frac{1}{3r^3} - \frac{1}{24} \\ \lim_{r \rightarrow 0^-} \int_{-2}^r x^{-4} dx &= \lim_{r \rightarrow 0^-} \left(-\frac{1}{3r^3} - \frac{1}{24} \right) \end{aligned}$$

which doesn't exist, so the integral cannot be evaluated.

2. Let $a_{n+1} = (a_n + 7)/2$, $a_1 = 19$. Prove that $a_n \rightarrow 7$.

Solution. First we show a_n is bounded below by 7. Certainly $a_1 > 7$, and if $a_n > 7$ then $a_{n+1} = (a_n + 7)/2 > (7 + 7)/2 = 7$, so by induction $a_n > 7$ for all n .

Next we show a_n is decreasing. Since $a_n > 7$ we have

$$a_{n+1} = (a_n + 7)/2 < (a_n + a_n)/2 = a_n$$

so a_n is decreasing.

It follows that $\lim_{n \rightarrow \infty} a_n$ exists. Call the limit L . Then

$$\lim_{n \rightarrow \infty} a_{n+1} = \lim_{n \rightarrow \infty} (a_n + 7)/2 = ((\lim_{n \rightarrow \infty} a_n) + 7)/2$$

so $L = (L + 7)/2$, so $L = 7$.

3. Prove that $\sum_{n=1}^{\infty} (-1)^{n-1} n^{-1/3}$ is conditionally convergent and not absolutely convergent.

Solution. Let $a_n = n^{-1/3} = 1/\sqrt[3]{n}$. Then $a_n > 0$, a_n is decreasing, and $a_n \rightarrow 0$, so by the alternating series test $\sum_{n=1}^{\infty} (-1)^{n-1} n^{-1/3}$ converges.

However, $\sum_{n=1}^{\infty} |(-1)^{n-1} n^{-1/3}| = \sum_{n=1}^{\infty} n^{-1/3}$ is a p -series with $p = 1/3 \leq 1$ so it diverges.

Therefore the original series converges conditionally and not absolutely.

Test A Question 4.

Determine the values of k such that the following linear system has:

- (i) a unique solution; (ii) no solution; (iii) more than one solution.

$$kx + y + z = 1$$

$$x + ky + z = 1$$

$$x + y + kz = 1$$

Solution

$$\begin{array}{l} \left[\begin{array}{ccc|c} k & 1 & 1 & 1 \\ 1 & k & 1 & 1 \\ 1 & 1 & k & 1 \end{array} \right] \quad \begin{array}{l} R3 \\ R2 - R3 \\ R1 - kR3 \end{array} \left[\begin{array}{ccc|c} 1 & 1 & k & 1 \\ 0 & k-1 & 1-k & 0 \\ 0 & 1-k & 1-k^2 & 1-k \end{array} \right] \end{array}$$

We see from this that there are infinitely many solutions when $k = 1$.

$$\begin{array}{l} \text{Assume } k \neq 1. \quad \begin{array}{l} R1 \\ R2 / k-1 \\ R3 / k-1 \end{array} \left[\begin{array}{ccc|c} 1 & 1 & k & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & 1+k & 1 \end{array} \right] \quad \begin{array}{l} R1 - R2 \\ R2 \\ R3 - R2 \end{array} \left[\begin{array}{ccc|c} 1 & 0 & k+1 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & k+2 & 1 \end{array} \right] \end{array}$$

We see from this that there are no solutions for $k = -2$ and a unique solution for $k \neq 1, k \neq -2$. The answers are therefore:

- (i) $k \neq 1, k \neq -2$; (ii) $k = -2$; (iii) $k = 1$.

Test A Question 5. Find the inverse of the matrix $\begin{bmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{bmatrix}$.

Solution

$$\begin{array}{l} \left[\begin{array}{ccc|ccc} -1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 & 1 & 0 \\ 4 & -2 & 5 & 0 & 0 & 1 \end{array} \right] \quad \begin{array}{l} -R1 \\ R2 + 2R1 \\ R3 + 4R1 \end{array} \left[\begin{array}{ccc|ccc} 1 & -2 & -3 & -1 & 0 & 0 \\ 0 & 5 & -6 & 2 & 1 & 0 \\ 0 & 6 & -7 & 4 & 0 & 1 \end{array} \right] \quad \begin{array}{l} R1 \\ R3 - R2 \\ R2 \end{array} \left[\begin{array}{ccc|ccc} 1 & -2 & -3 & -1 & 0 & 0 \\ 0 & 1 & -1 & 2 & -1 & 1 \\ 0 & 5 & -6 & 2 & 1 & 0 \end{array} \right] \\ \\ \begin{array}{l} R1 + 2R2 \\ R2 \\ R3 - 5R2 \end{array} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 3 & -2 & 2 \\ 0 & 1 & -1 & 2 & -1 & 1 \\ 0 & 0 & -1 & -8 & 6 & -5 \end{array} \right] \quad \begin{array}{l} R1 + R3 \\ R2 - 3 \\ -R3 \end{array} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -5 & 4 & -3 \\ 0 & 1 & 0 & 10 & -7 & 6 \\ 0 & 0 & 1 & 8 & -6 & 5 \end{array} \right] \end{array}$$

The required inverse is $\begin{bmatrix} -5 & 4 & -3 \\ 10 & -7 & 6 \\ 8 & -6 & 5 \end{bmatrix}$.

Test A Question 6. Find three values of x for which $\begin{vmatrix} 1 & 2x & x^2 & x^3 \\ 1 & -2 & 1 & -1 \\ 1 & 4 & 4 & 8 \\ 1 & 6 & 9 & 27 \end{vmatrix} = 0$.

Solution

The values are $x = -1, 2$ and 3 since each of these values will cause two rows of the determinant to be equal.

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1. Evaluate $\int_{-4}^5 x^{-6} dx$ or show that it cannot be evaluated.

Solution.

$$\int_{-4}^5 x^{-6} dx = \lim_{r \rightarrow 0^-} \int_{-4}^r x^{-6} dx + \lim_{s \rightarrow 0^+} \int_s^5 x^{-6} dx$$

if the limits exist.

$$\begin{aligned} \int_{-4}^r x^{-6} dx &= \left[-\frac{1}{5x^5} \right]_{-4}^r = -\frac{1}{5r^5} - \frac{1}{5120} \\ \lim_{r \rightarrow 0^-} \int_{-4}^r x^{-6} dx &= \lim_{r \rightarrow 0^-} \left(-\frac{1}{5r^5} - \frac{1}{5120} \right) \end{aligned}$$

which doesn't exist, so the integral cannot be evaluated.

2. Let $a_{n+1} = \sqrt{5a_n}$, $a_1 = 13$. Prove that $a_n \rightarrow 5$.

Solution. First we show a_n is bounded below by 5. Certainly $a_1 > 5$, and if $a_n > 5$ then $a_{n+1} = \sqrt{5a_n} > \sqrt{(5)(5)} = 5$, so by induction $a_n > 5$ for all n .

Next we show a_n is decreasing. Since $a_n > 5$ we have

$$a_{n+1} = \sqrt{5a_n} < \sqrt{a_n a_n} = a_n$$

so a_n is decreasing.

It follows that $\lim_{n \rightarrow \infty} a_n$ exists. Call the limit L . Then

$$\lim_{n \rightarrow \infty} a_{n+1} = \lim_{n \rightarrow \infty} \sqrt{5a_n} = \sqrt{5 \lim_{n \rightarrow \infty} a_n}$$

so $L = \sqrt{5L}$, so $L = 0$ or $L = 5$. Since $a_n > 5$ for all n , we reject $L = 0$ as impossible. Thus $L = 5$.

3. Prove that $\sum_{n=1}^{\infty} (-1)^{n-1} n^{-1/2}$ is conditionally convergent and not absolutely convergent.

Solution. Let $a_n = n^{-1/2} = 1/\sqrt{n}$. Then $a_n > 0$, a_n is decreasing, and $a_n \rightarrow 0$, so by the alternating series test $\sum_{n=1}^{\infty} (-1)^{n-1} n^{-1/2}$ converges.

However, $\sum_{n=1}^{\infty} |(-1)^{n-1} n^{-1/2}| = \sum_{n=1}^{\infty} n^{-1/2}$ is a p -series with $p = 1/2 \leq 1$ so it diverges.

Therefore the original series converges conditionally and not absolutely.

Test B Question 4. Determine the values of k such that the following linear system has:

- (i) a unique solution; (ii) no solution; (iii) more than one solution.

$$x - 3z = -3$$

$$2x + ky - z = -2$$

$$x + 2y + kz = 1$$

Solution

$$\begin{bmatrix} 1 & 0 & -3 & -3 \\ 2 & k & -1 & -2 \\ 1 & 2 & k & 1 \end{bmatrix} \quad \begin{array}{l} R1 \\ R2 - 2R1 \\ R3 - R1 \end{array} \begin{bmatrix} 1 & 0 & -3 & -3 \\ 0 & k & 5 & 4 \\ 0 & 2 & k+3 & 4 \end{bmatrix} \quad \begin{array}{l} R1 \\ R3 \div 2 \\ R2 \end{array} \begin{bmatrix} 1 & 0 & -3 & -3 \\ 0 & 1 & \frac{1}{2}(k+3) & 2 \\ 0 & k & 5 & 4 \end{bmatrix}$$

$$\begin{array}{l} R1 \\ R2 \\ R3 - kR2 \end{array} \begin{bmatrix} 1 & 0 & -3 & -3 \\ 0 & 1 & \frac{1}{2}(k+3) & 2 \\ 0 & 0 & \frac{1}{2}(k+5)(2-k) & 2(2-k) \end{bmatrix}$$

If $k = 2$, we obtain $\begin{bmatrix} 1 & 0 & -3 & -3 \\ 0 & 1 & \frac{5}{2} & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$; so the system has many solutions.

If $k = -5$, we obtain $\begin{bmatrix} 1 & 0 & -3 & -3 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 14 \end{bmatrix}$; so the system has no solution.

The answers are therefore:

- (i) $k \neq -5$ and $k \neq 2$; (ii) $k = -5$; (iii) $k = 2$.

Test B Question 5. Find the inverse of the matrix $\begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 5 & 2 & -3 \end{bmatrix}$.

Solution

$$\begin{bmatrix} 2 & 1 & -1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 5 & 2 & -3 & 0 & 0 & 1 \end{bmatrix} \quad \begin{array}{l} R1 \div 2 \\ R2 \div 2 \\ R3 - \frac{5}{2}R1 \end{array} \begin{bmatrix} 1 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 1 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & -\frac{1}{2} & -\frac{5}{2} & 0 & 1 \end{bmatrix} \quad \begin{array}{l} R1 - \frac{1}{2}R2 \\ R2 \\ R3 + \frac{1}{2}R2 \end{array} \begin{bmatrix} 1 & 0 & -\frac{3}{4} & \frac{1}{2} & -\frac{1}{4} & 0 \\ 0 & 1 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & -\frac{1}{4} & -\frac{5}{2} & \frac{1}{4} & 1 \end{bmatrix}$$

$$\begin{array}{l} R1 \\ R2 \\ -4R3 \end{array} \begin{bmatrix} 1 & 0 & -\frac{3}{4} & \frac{1}{2} & -\frac{1}{4} & 0 \\ 0 & 1 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 & 10 & -1 & -4 \end{bmatrix} \quad \begin{array}{l} R1 + \frac{3}{4}R3 \\ R2 - \frac{1}{2}R3 \\ R3 \end{array} \begin{bmatrix} 1 & 0 & 0 & 8 & -1 & -3 \\ 0 & 1 & 0 & -5 & 1 & 2 \\ 0 & 0 & 1 & 10 & -1 & -4 \end{bmatrix}$$

So the required inverse is $\begin{bmatrix} 8 & -1 & -3 \\ -5 & 1 & 2 \\ 10 & -1 & -4 \end{bmatrix}$.

Test B Question 6. Find three values of x for which $\begin{vmatrix} x^3 & 2x^2 & 3x & 1 \\ 8 & 8 & 6 & 1 \\ -1 & 2 & -3 & 1 \\ -27 & 18 & -9 & 1 \end{vmatrix} = 0$.

Solution The values are $x = 2, -1$ and -3 since each of these values will cause two rows of the determinant to be equal.